ONE-PAGE BRIEF

Post-quantum cryptography review: hyperledger-fabric @ 13ef9e5ee594

Client: Hyperledger Fabric (LF Decentralized Trust) — public repository (passive review) · Source: github.com/hyperledger/fabric, commit `13ef9e5ee594` · Reviewed: 2026-11-06 10:32 UTC · Risk profile: Crypto exchange, custodian or wallet

POST-QUANTUM MATURITY

Level 4 of 4

post-quantum by default in the critical paths

FINDINGS

0 Critical **1** High **7** Medium **3** Low **18** Info

QUANTUM TIMELINE (MOSCA: X + Y − Z)

Key establishment: 10 + 5 − 8 = 7 → **too late**

Signatures: 10 + 5 − 8 = 7 → **too late**

Next step: Keep the inventory current and review new code paths before release.

WHAT MATTERS MOST

High	QSI-2026-17550	**google.golang.org/grpc v1.83.1 is affected by GHSA-2v4p-qf9q-27wj (CVE-2026-84445, GO-2026-6443)** What to do: Upgrade to a fixed version (1.82.2, 1.83.2, 1.84.0-dev.0.20260825144003-d5a41119e0e3, 1.85.0-dev.0.20260825072537-93e31b48545e).
Medium	QSI-2026-17522	**Quantum-vulnerable ECDSA: transaction and message signing (1 file(s))** preliminary What to do: Follow each chain's post-quantum roadmap, avoid public-key reuse, and keep signers ready to add post-quantum signature schemes as chains...
Medium	QSI-2026-17523	**Quantum-vulnerable Ed25519: transaction and message signing (1 file(s))** preliminary What to do: Follow each chain's post-quantum roadmap, avoid public-key reuse, and keep signers ready to add post-quantum signature schemes as chains...
Medium	QSI-2026-17524	**Quantum-vulnerable ECDSA: role not determined (8 file(s))** preliminary What to do: Classify each usage, record data lifetimes, and plan migration to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form...

7 more in the full report.

WHAT THE QUANTUM-VULNERABLE CRYPTOGRAPHY PROTECTS

Role	Algorithms	What to do
Medium transaction and message signing	ECDSA, Ed25519	Follow each chain's post-quantum roadmap, avoid public-key reuse, and keep signers ready to add post-quantum...
Medium authentication and identity (certificates, tokens, device attestation)	ECDSA	Keep credentials short-lived and plan ML-DSA certificates, tokens and keys as the ecosystem supports them
Medium role not determined	ECDSA, ECDH, Ed25519	Classify each usage, record data lifetimes, and plan migration to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205)...

KNOWN VULNERABLE COMPONENTS

Advisories for the versions in use: 20; unlikely to apply to this build: 18 (a Cargo feature that is off, affected functions not named in the code, or a platform that is not built); to review: 2.

Static review of source code and manifests; files analysed: 814. No requests were sent to the owner's systems.

Full report, content SHA-256: `c6d8ad0dc0ae1f4057b7a287b056a05648ac2555beac30700886e6ad50e9e186`

Full report signature (report.sig.json): Ed25519 key, SHA-256 fingerprint `5b3cf9c9bd0fa54447e2b5dfa7a6bab569bd0c666fa46949599e5c348d17083e`. bundle.sig.json lists the SHA-256 of every file in this bundle and is signed too; verify-bundle checks it offline.

Preliminary findings need confirmation in the code; the full report gives the evidence, the limits of each check and how to re-test.